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(54) FLEXIBLE HIGH-POWER RF LINE WITH JOINTS

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(57) ABSTRACT

A tiltable RF joint for high power coaxial RF lines includes a first outer conductor with a first inner conductor centered therein and a second outer conductor with a second inner conductor centered therein. The first inner conductor that contains a joint socket is mechanically coupled to a joint ball of the second inner conductor and forms a socket-ball connection. The first inner conductor also contains a center contact surface that is shaped as a spherical segment and that is in contact with a center contact spring of the second inner conductor. Any one of the outer conductors has an outer conductor contact spring that is in contact with an outer conductor contact surface at the other outer conductor.

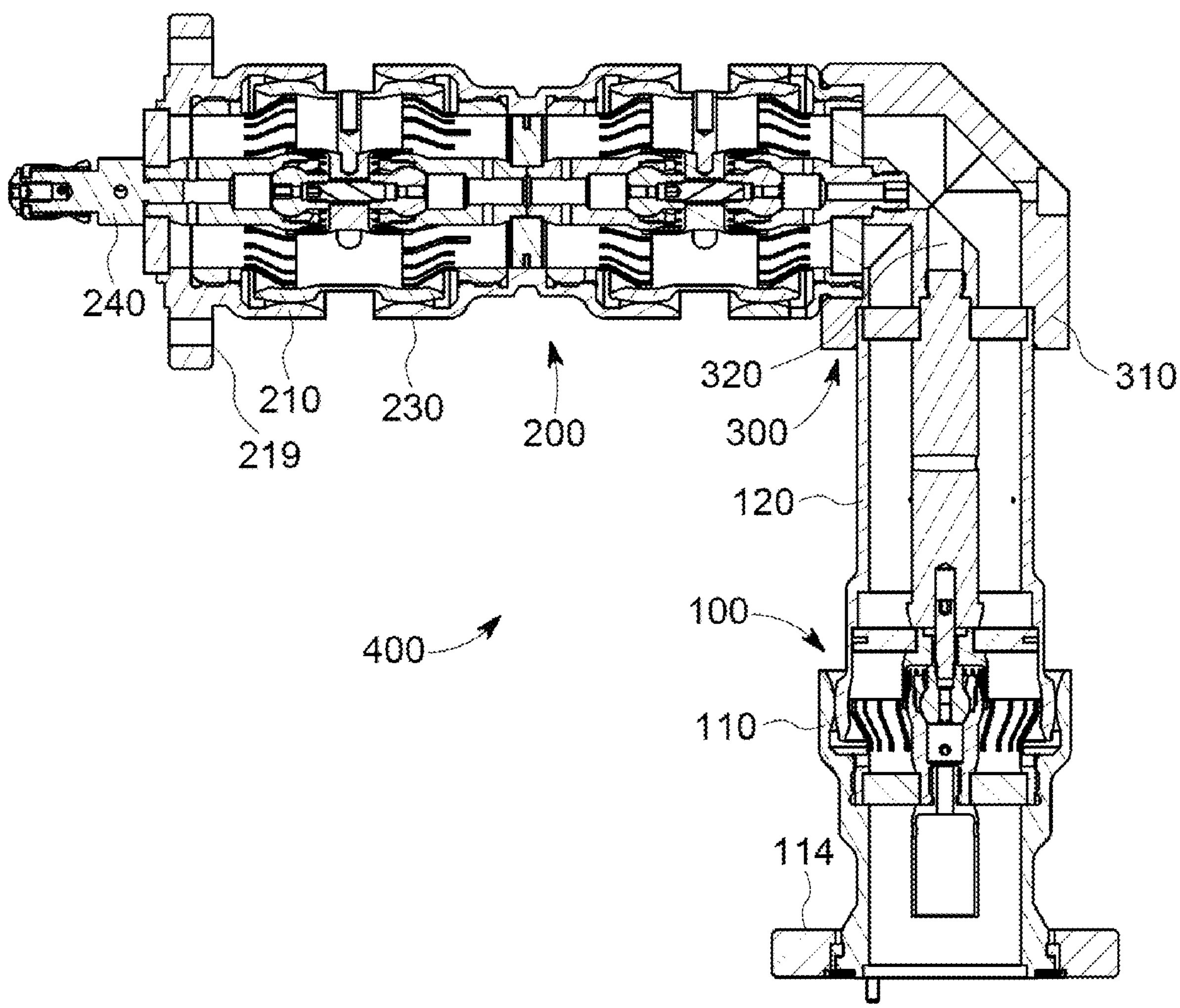

Fig. 1

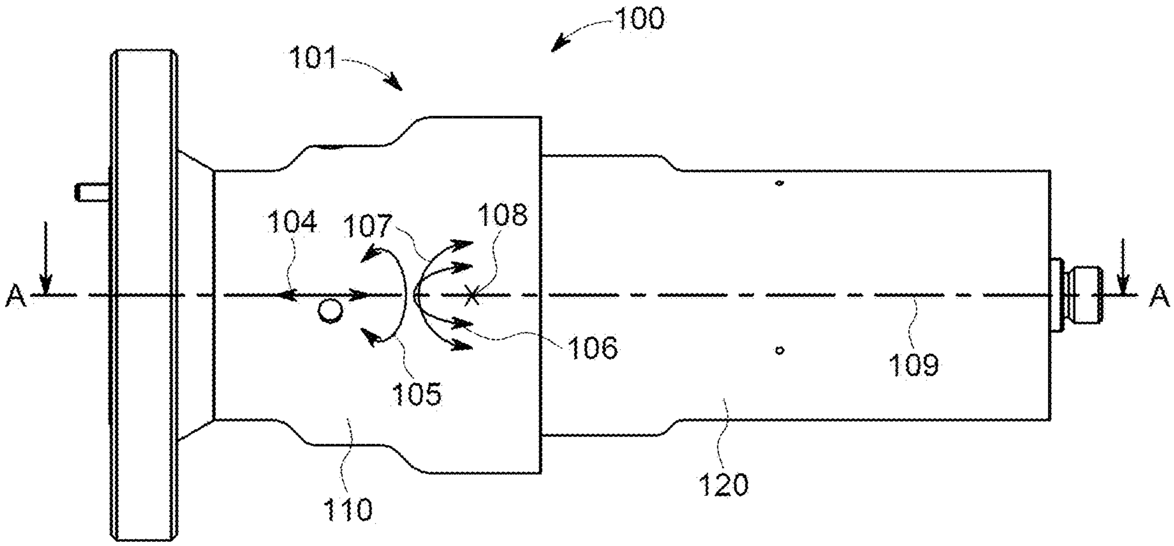
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Fig. 2

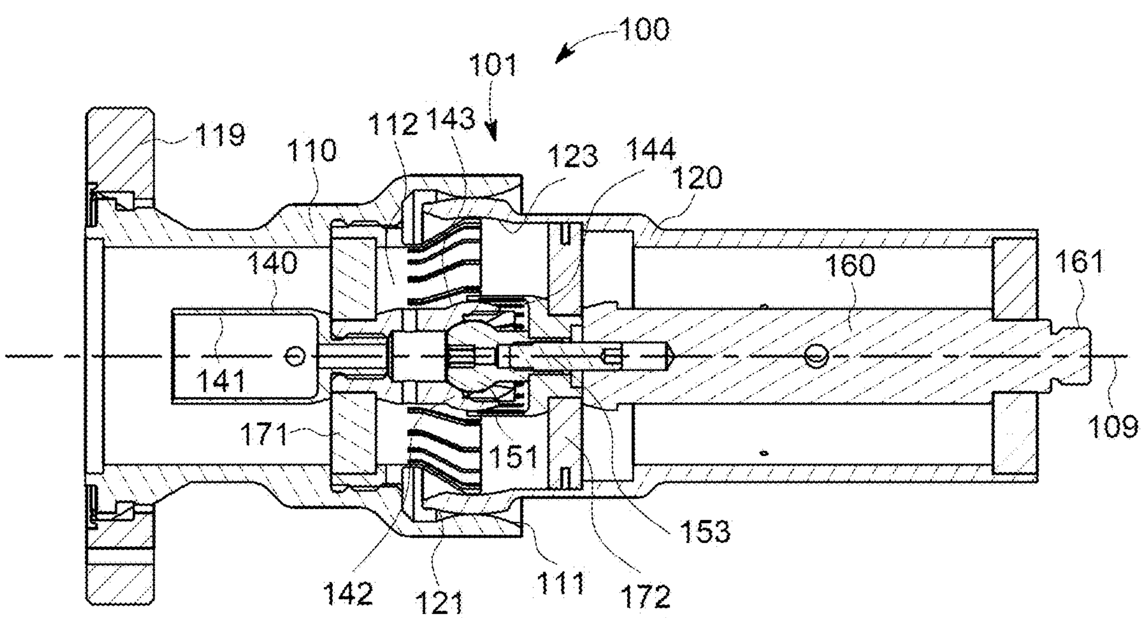
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Fig. 3

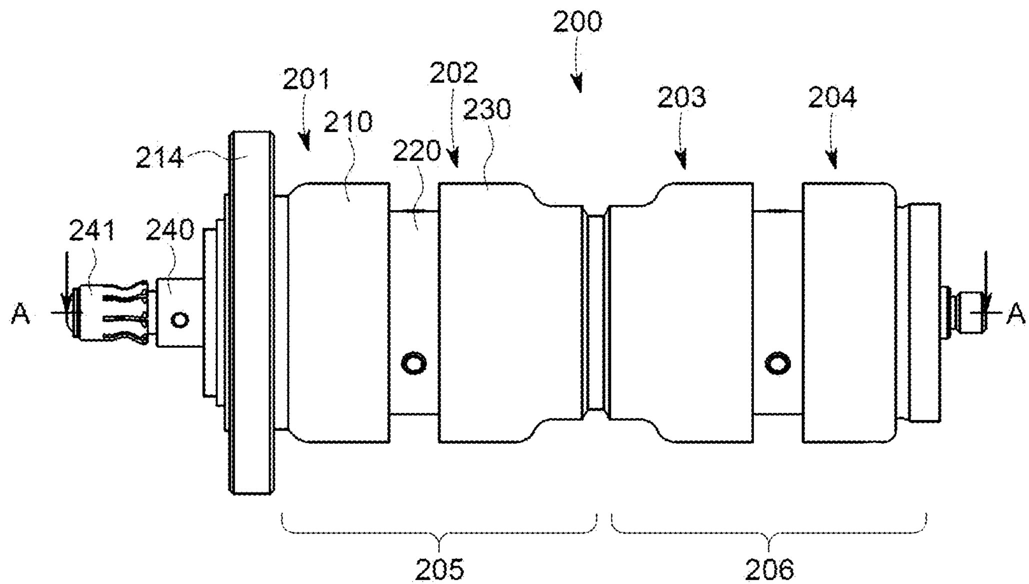
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Fig. 4

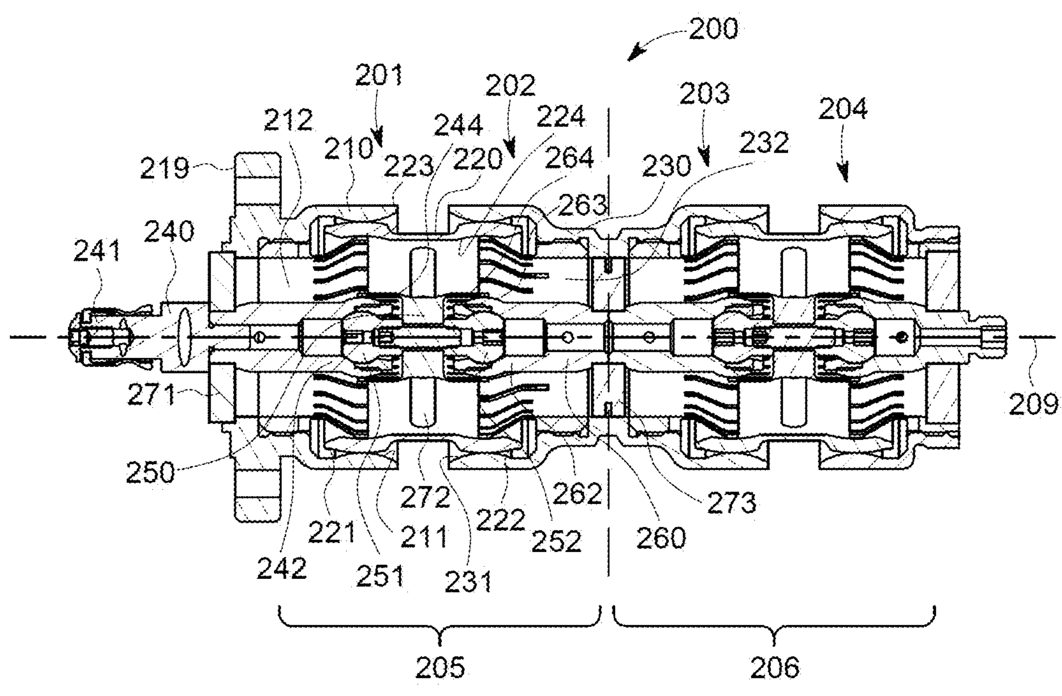
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Fig. 5

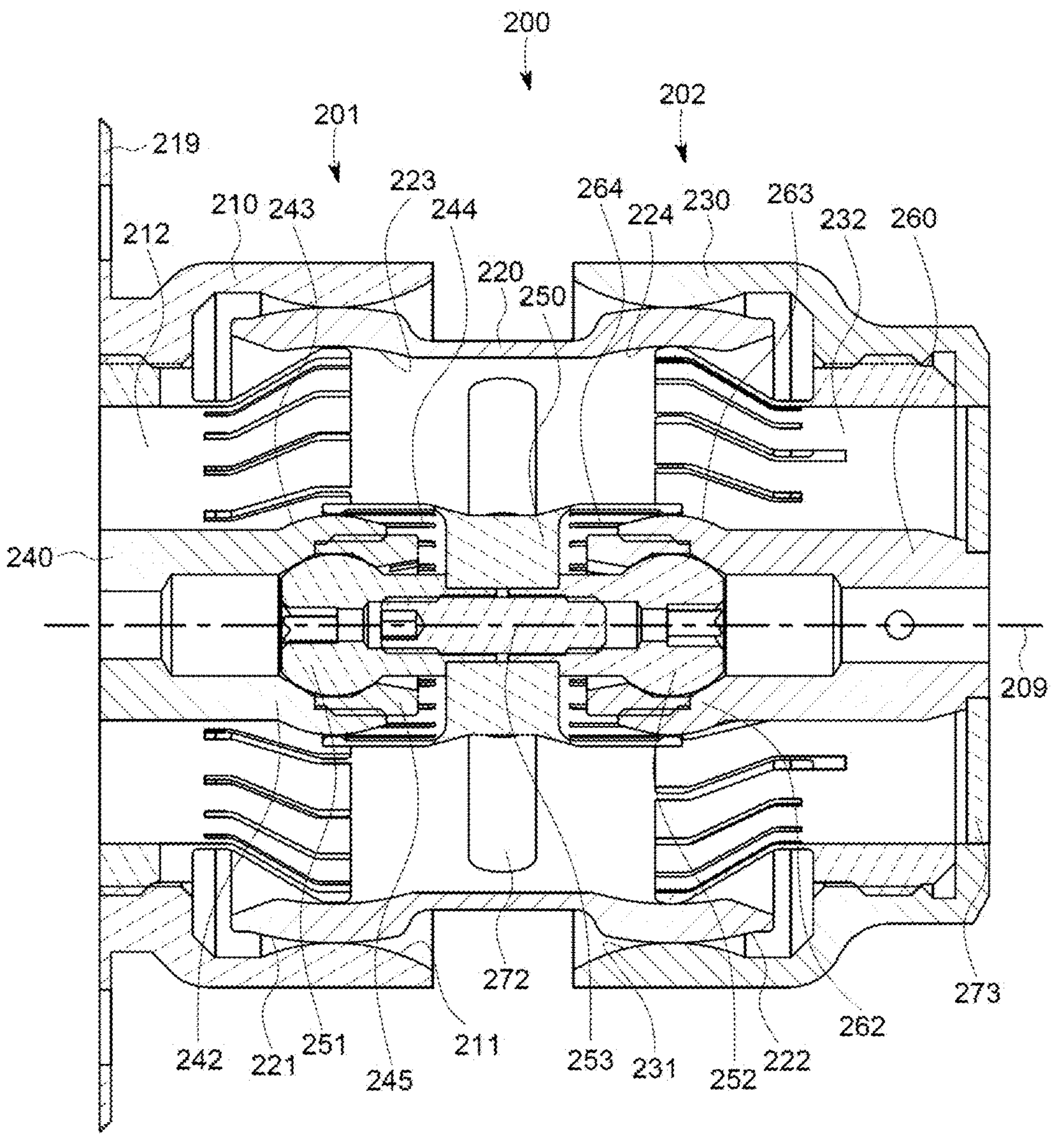
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Fig. 6

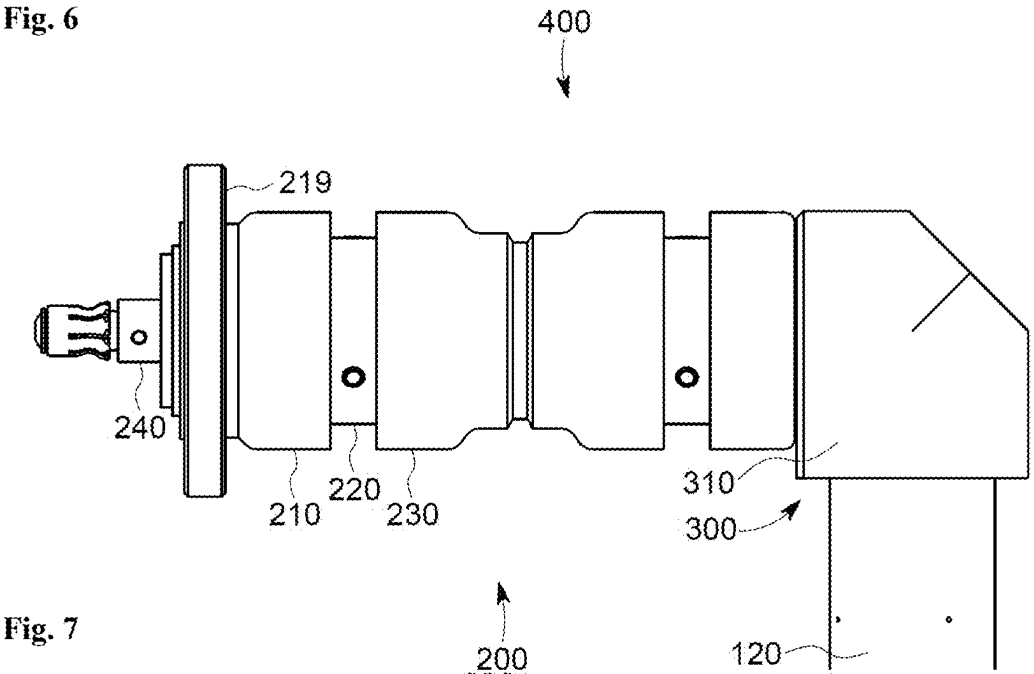

Fig. 7

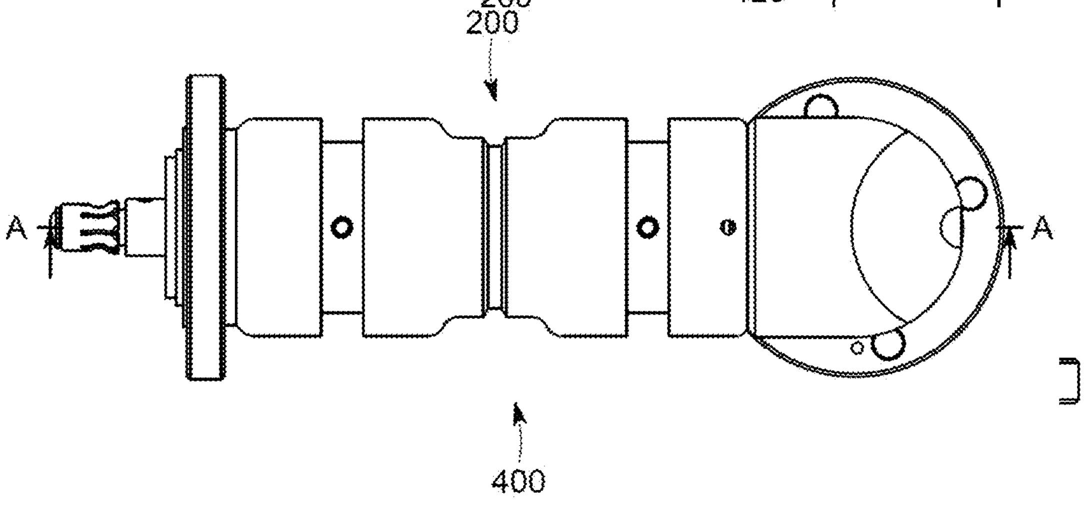

Fig. 8

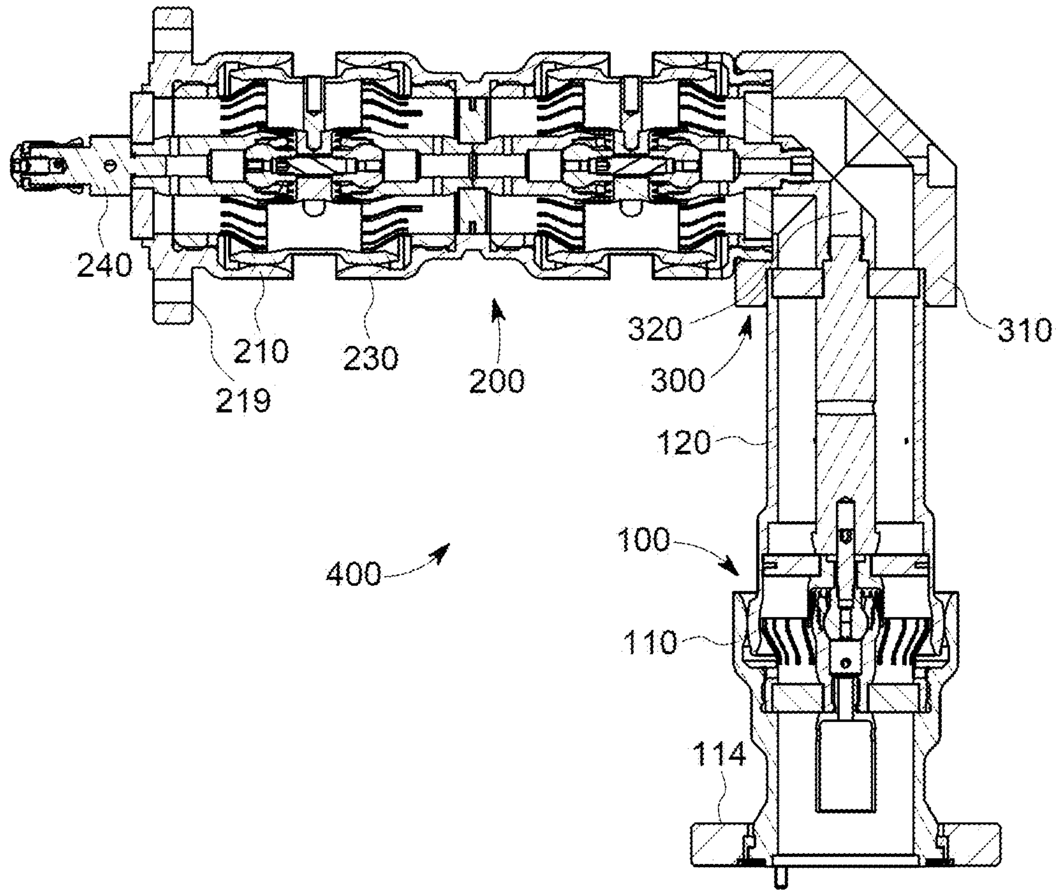
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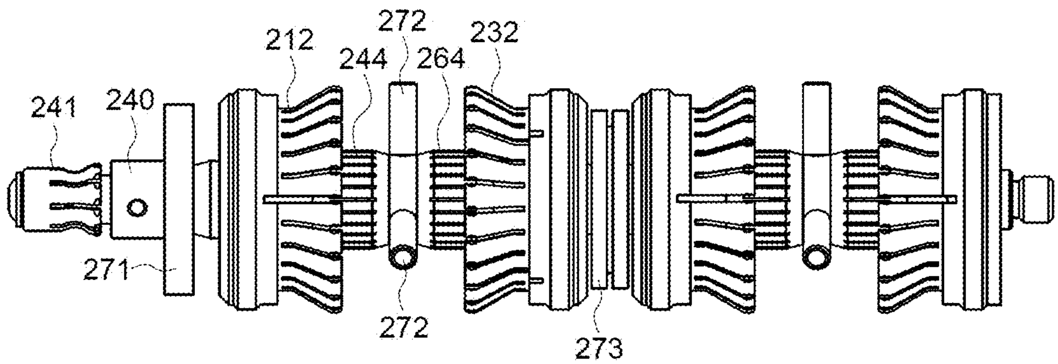
Fig. 9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FLEXIBLE HIGH-POWER RF LINE WITH JOINTS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This US Patent application claims priority to the pending European Application No. 24160900.7 filed on Mar. 1, 2024, the disclosure of which is incorporated herein by reference.

BACKGROUND

1. Field of the Invention

[0002] The invention relates to tiltable and/or rotatable joints for high power coaxial RF lines and flexible high power RF lines with joints.

2. Description of the Related Art

[0003] For a connection between high power RF devices like power transmitters and loads sometimes a certain degree of flexibility is required. A coaxial RF connector system for connecting cables is disclosed in EP 3 300 535 A1. This connector system can couple comparatively high RF power up to a few Kilowatts. For higher power levels larger connectors and larger diameter cables are required. Such cables are comparatively inflexible and require high forces for disconnecting the connectors and removing the cables or components attached to the cables.

SUMMARY OF THE INVENTION

[0004] The embodiments are providing tiltable and/or rotatable joints for high power coaxial RF lines and an RF connecting system which is capable of transferring high power RF signals in the range of many Kilowatts and above in a broad frequency range between DC and 3.5 GHz, but not limited to that range.

[0005] In an embodiment, a tiltable and/or rotatable RF joint for high power coaxial RF lines includes a first outer conductor and a second outer conductor. A first inner conductor is centered within the first outer conductor and has a first center axis. A second inner conductor is centered within the second outer conductor and has a second center axis. The first center axis and the second center axis may identify or define the same center axis of the tiltable RF joint when the tiltable RF joint is straight (not tilted). The first inner conductor has a joint socket mechanically coupled to a joint ball of the second inner conductor, forming a socket-ball connection. This socket-ball connection may allow a tilt or rotation in at least one axis or in two axes orthogonal to each other and relative to the corresponding center axis of at least one of the inner conductors. A first tilt may be in a plane through said center axis of the tiltable RF joint while a second tilt may be orthogonal to said plane. Further, there may be a rotation around said center axis of the tiltable RF joint. The socket-ball connection may further allow a limited axial movement. There may be any combination of the directions of tilt, rotation and linear displacement. This kind of tiltable/rotatable RF joint may also be referred to as a ball joint. Herein, for simplicity, reference is made to a tiltable joint, which means the tiltable, rotatable, displaceable joint mentioned above. The socket may have a locking means, which may hold the ball at the socket and/or restrict axial movement.

[0006] Generally, the RF joint is formed with respect to a center axis (of such RF joint) that is defined by either a first inner conductor or a second inner conductor configured for at least one of:

- **[0007]** tilt in a plane through the center axis of the RF joint,
- **[0008]** tilt orthogonal to the plane through the center axis of the RF joint,
- **[0009]** rotation around the center axis of the RF joint,
- **[0010]** extension or reduction in length along the center axis of the RF joint.

[0011] While the socket-ball connection may have a mechanical function, electrical contact means are provided for inner conductor contacting. The first inner conductor may have a center contact surface further having the shape of a spherical segment, which is in contact with a center contact spring at the second inner conductor. The center contact spring may be a hollow body which may further include a plurality of slots. It may have a hollow cylindrical or spherical shape or a segment thereof. It may touch the center contact surface from its outside. The center contact surface may be at the outside of the joint socket. The center contact surface may have the shape of a spherical segment with a center point at the center point of the joint ball. This provides a constant radius of the spring and therefore a constant spring force over tilting.

[0012] To provide an electrical contact between the outer conductors, the first outer conductor may have an outer conductor contact spring which is in contact with an outer conductor contact surface at the second outer conductor. This may also be reversed, such that the second outer conductor may have an outer conductor contact spring which is in contact with an outer conductor contact surface at the first outer conductor. The outer conductor contact spring may be a hollow body. It may have a hollow cylindrical or spherical shape. It may touch the outer conductor contact surface from its inner side. The outer conductor contact surface may have a hollow spherical shape or a segment thereof, which may have a center point at the center point of the joint ball. This provides a constant radius of the spring and therefore a constant spring force over tilting.

[0013] To enclose the tiltable RF joint and protect its interior from the environment, the outer conductors may overlap each other. The second outer conductor may have an outer bearing surface, which may have the shape of a spherical segment. This may be overlapped by an inner bearing surface of the first outer conductor. This order may also be reversed, such that the first outer conductor may have an outer bearing surface, which may have the shape of a spherical segment, and this may be overlapped by an inner bearing surface of the second outer conductor. The overlapping surface may also have a shape of a spherical segment. The outer bearing surface may have a convex shape while the inner bearing surface may either have a concave or a convex surface.

[0014] To hold the inner conductors in place at a center position within the outer conductors, at least one spacer may be provided between an inner conductor and its outer conductor. There may be a first spacer between the first inner conductor and the first outer conductor. Further, a second spacer may be provided between the second inner conductor and the second outer conductor.

[0015] The angle of tilt of a tiltable RF joint may be restricted in any direction withing a range of ±60° or less, e.g. ±45° or ±30°. There is no need for restricting the angle of rotation around the center axis, but it may be restricted to 360° or less, 180° or less or 90° or less. A linear displacement may be restricted to 20 mm or less, 10 mm or less, or 5 mm or less. It may also be restricted to a values smaller than 1 mm. The inner conductors may have a diameter in a range from 5 mm to 500 mm or 10 mm to 50 mm. The outer conductors may have a diameter between 2 to 5 times larger than the inner conductors.

[0016] A joint component may include a tiltable RF joint as described above. It may further include electrical contacting means, e.g. inner conductor contacts and/or mechanical attachment means, e.g. a connection flange and/or at least one elongated conductor section including corresponding outer and inner conductors. The inner conductor contacts may be pins or sockets. They may include at least one spring.

[0017] In an embodiment, a joint component may include multiple tiltable RF joints connected in series. This allows for a higher degree in mechanical flexibility. There may be 2, 3, 4 or more such tiltable RF joint connected together.

[0018] In a specific embodiment, a joint component may include two tiltable RF joints connected in series and mirrored to each other such that e.g. both second inner conductors are connected together. This has the benefit, that the joint balls of both tiltable RF joints can be held by a common joint ball locking means, which simplifies assembly. Multiples of such joint components may be connected together for further increasing flexibility.

[0019] An RF connecting system may include a plurality of joint components connected together. Further, at least one of a straight, curved or cornered conductor component may be added. Such a conductor component may include an outer conductor and an inne conductor centered within the outer conductor. A cornered conductor component, also called elbow may have a 90° angle, but other angles e.g. 60° or 45° or any value in between are possible.

BRIEF DESCRIPTION OF THE DRAWINGS

[0020] In the following, the invention will be described by way of example, without limitation of the general inventive concept, on examples of embodiment and with reference to the drawings.

[0021] FIG. **1** shows a line section with a single tiltable RF joint.

[0022] FIG. **2** shows a sectional view of the line section with a single tiltable RF joint.

[0023] FIG. **3** shows a line section with two dual tiltable RF joints.

[0024] FIG. **4** shows a sectional view of the line section with two tiltable RF joints.

[0025] FIG. **5** shows an enlarged section of FIG. **4**.

[0026] FIG. **6** shows a side view of a RF connecting system.

[0027] FIG. **7** shows a sectional view of a RF connecting system.

[0028] FIG. **8** shows a top view of a RF connecting system.

[0029] FIG. **9** shows the contact springs of the second tiltable RF joint.

[0030] Generally, the drawings are not to scale. Like elements and components are referred to by like labels and numerals. For the simplicity of illustrations, not all elements and components depicted and labeled in one drawing are necessarily labels in another drawing even if these elements and components appear in such other drawing.

[0031] While various modifications and alternative forms, of implementation of the idea of the invention are within the scope of the invention, specific embodiments thereof are shown by way of example in the drawings and are described below in detail. It should be understood, however, that the drawings and related detailed description are not intended to limit the implementation of the idea of the invention to the particular form disclosed in this application, but on the contrary, the intention is to cover all modifications, equivalents and alternatives falling within the spirit and scope of the present invention as defined by the appended claims.

DETAILED DESCRIPTION

[0032] FIG. **1** shows an embodiment of a first joint component with a single ball joint. The first joint component **100** includes a tiltable RF joint **101**. The first joint component **100** has a first joint center axis **109** and a center of ball joint **108** on the first joint center axis **109**. A first section, e.g., comprising first outer conductor **110** may be tilted against a second section, e.g., comprising second outer conductor **120**. There may a be a first direction of tilt, orthogonal to the drawing plane **106** and a second direction of tilt in the drawing plane **107**. Herein tilt may be a rotational movement within a restricted range of angles. Further, there may be a rotation **105** around the first joint center axis **109** and/or a linear displacement **104** along the first joint center axis **109**. There may be any combination of the directions of tilt, rotation and linear displacement. Generally, each of the joint components may have its own center axis. In the embodiment shown, the joint component forms a straight line such that both center axis is the same.

[0033] FIG. **2** shows a sectional view of the first joint component **100** of FIG. **1**. Its first section includes a first outer conductor **110** having an inner bearing surface **111** which may be in mechanical contact with an outer bearing surface **121** of the second section which includes a second outer conductor **120**. Such a mechanical contact is not required but may increase stability and can help sealing of the inner components. The first section further includes an outer conductor contact spring **112** forming an electrical contact with an outer conductor surface **123** of the second section. This establishes a continuous electrical connection between the outer conductors of the first section and the second section.

[0034] The first section further includes a first inner conductor **140** having a first inner conductor contact **141** for external contacting the joint section. The first inner conductor **140** further includes a joint socket **142** which is in mechanical contact with a joint ball **151** of the second section forming a socket-ball connection which basically allows tilting between the sections. A center contact surface **143** at the first section is contacted by a center contact spring **144** of the second section. The joint ball **151** may be locked by a joint ball locking means **153** which may be a screw and allows easy assembly and disassembly of the sections.

[0035] In the second section, a second inner conductor **160** having a second inner conductor contact **161** may be connected to the joint ball **151** and the center contact spring **144**, which may include a plurality of slots.

[0036] For holding the inner conductors in place at a center position within the outer conductors, a first spacer **171** may be provided between the first inner conductor **140** and the first outer conductor **110**. Further, a second spacer **172** and a third spacer **173** may be provided between the second inner conductor **160** and the second outer conductor **120**.

[0037] For mechanically mounting and connecting of the outer conductors, a first connection flange **119** may be provided at the first outer conductor **110**.

[0038] FIG. **3** shows a line section with two dual tiltable RF joints. The second joint component **200** includes a first tiltable RF joint **201**, a second tiltable RF joint **202**, a third tiltable RF joint **203** and a fourth tiltable RF joint **204**. Basically, all these tiltable RF joint are configured like the tiltable RF joint **101** of first joint component **100**. The first tiltable RF joint **201** and second tiltable RF joint **202** form a first joint assembly **205**. The third tiltable RF joint **203** and fourth tiltable RF joint **204** form a second joint assembly **206**. As the second joint assembly **206** is basically identical to the first joint assembly **205**, herein only the first joint assembly **205** is described in detail. The second joint component **200** has a second joint center axis **209**.

[0039] The second joint component is a dual joint component which basically includes two joints of the first joint component in a mirrored arrangement. For better readability, the parts of the first joint are denoted by "A" whereas the parts of the second (mirrored) joint are denoted as "B". The second joint component has a first inner conductor A **240** further having a first inner conductor A contact **241**. A second connection flange **219** may be provided for electrically and mechanically connecting the second joint component **200**. For holding the inner conductors in place relative to the outer conductors, a first spacer **271** may be between first inner conductor A **240** and first outer conductor A **210**. There may be second spacers **272** between a dual second inner conductor **250** and a dual second outer conductor **220**. Further, there may be a third spacer **273** between first inner conductor B **260** and first outer conductor B **230**.

[0040] Further details are described in an enlarged view of the next figure.

[0041] FIG. **5** shows an enlarged section of FIG. **4**. Here, details of the first tiltable RF joint **201** and the second tiltable RF joint **202** are shown. The second tiltable RF joint **202** is basically mirrored to the first tiltable RF joint **201**. A first outer conductor A **210** has an inner bearing surface A **211** which may be in contact with an outer bearing surface A **221** of a dual second outer conductor **220**. An outer conductor contact spring A **212** may be in mechanical and electrical contact with outer conductor contact surface A **223** of dual second outer conductor **220**. The first inner conductor A **240** may include a joint socket **242** that is forming a socket ball connection with joint ball A **251** being part of a dual second inner conductor **250**. This socket ball connection primarily has a mechanical function and defines the center of tilt or rotation. The socket may have an elongated shape, such that the ball may also be displaced in an axial direction. The socket may have a locking means **245**, which may hold the ball at the socket and/or restrict axial movement. For electrical connection of the inner conductors, the first inner conductor A **240** has a center contact surface A **243** which may be in electrical and mechanical contact with a center contact spring A **244**.

[0042] The second tiltable RF joint **202** in this embodiment is basically a mirrored version of the first tiltable RF joint **201**. The second tiltable RF joint **202** has a first outer conductor B **230** forming an inner bearing surface B **231** which may be in mechanical contact with an outer bearing surface B **222** of the dual second inner conductor **250**. The second tiltable RF joint **202** may further include an outer conductor contact spring B **232** which may be in electrical and mechanical contact with an outer conductor contact surface B **224** of the dual second inner conductor **250**. A socket ball connection is defined by a joint socket B **262** being part of the first inner conductor B **260** and a joint ball B **252** being part of the dual second inner conductor **250**. The joint ball B **252** may be locked with the joint ball A **251** to the dual second inner conductor **250** by a second and joint ball locking means **253**. An electrical center contact is established by a center contact spring B **264** being part of the dual second inner conductor **250** which is further in contact with a center contact surface B **263** being part of the first inner conductor B **260**.

[0043] FIG. **6** shows a side view of an RF connecting system. An exemplary embodiment of a RF connecting system **400** includes a first joint component **100**, a second joint component **200** and a corner section **300** with a corner outer conductor **310**.

[0044] FIG. **7** shows a top view of an RF connecting system **400**.

[0045] FIG. **8** shows a sectional view of an RF connecting system **400**. In this figure, basically the internal components can be seen. Details of the internal components have been explained in the previous figures. Here, further a corner inner conductor **320** of the corner section **300** is shown.

[0046] In FIG. **9**, the contact springs of the second joint component **200** are shown in more detail. The outer conductor contact spring A **212** and the outer conductor contact spring B **232** are part of the first joint assembly **205**. They have outward directed spring elements, which are configured to press against the outer conductor contact surface A **223** and the outer conductor contact surface B **224**. Due to the outwardly extending shape, they can maintain an electrical contact even through a tilting movement. The center contact spring A **244** and the center contact spring B **264** are configured to contact the center contact surface A **243** and the center contact surface B **263**. Instead of a single spacer, there may be a plurality of second spacers **272** or spacer sections arranged between the center contact spring A **244** and the center contact spring B **264**, for holding dual second inner conductor **250** in a defined position within the dual second outer conductor **220**.

[0047] It will be appreciated to those skilled in the art having the benefit of this disclosure that this invention is believed to provide RF joints and components. Further modifications and alternative embodiments of various aspects of the invention will be apparent to those skilled in the art in view of this description. Accordingly, this description is to be construed as illustrative only and is provided for the purpose of teaching those skilled in the art the general manner of carrying out the invention. It is to be understood that the forms of the invention shown and described herein are to be taken as the presently preferred embodiments. Elements and materials may be substituted for those illustrated and described herein, parts and processes may be reversed, and certain features of the invention may be utilized independently, all as would be apparent to one skilled in the art after having the benefit of this description of the invention. Changes may be made in the elements described herein without departing from the spirit and scope of the invention as described in the following claims.

LIST OF REFERENCE NUMERALS

[0048] **100** first joint component
[0049] **101** tiltable RF joint
[0050] **104** direction of linear movement
[0051] **105** direction of rotation
[0052] **106** direction of tilt orthogonal to the drawing plane
[0053] **107** direction of tilt in the drawing plane
[0054] **108** center of ball joint
[0055] **109** first joint center axis
[0056] **110** first outer conductor
[0057] **111** inner bearing surface
[0058] **112** outer conductor contact spring
[0059] **119** first connection flange
[0060] **120** second outer conductor
[0061] **121** outer bearing surface
[0062] **123** outer conductor contact surface
[0063] **140** first inner conductor
[0064] **141** first inner conductor contact
[0065] **142** joint socket
[0066] **143** center contact surface
[0067] **144** center contact spring
[0068] **151** joint ball
[0069] **153** joint ball locking means
[0070] **160** second inner conductor
[0071] **161** second inner conductor contact
[0072] **171** first spacer
[0073] **172** second spacer
[0074] **173** third spacer
[0075] **200** second joint component
[0076] **201** first tiltable RF joint
[0077] **202** second tiltable RF joint
[0078] **203** third tiltable RF joint
[0079] **204** fourth tiltable RF joint
[0080] **205** first joint assembly
[0081] **206** second joint assembly
[0082] **209** second joint center axis
[0083] **210** first outer conductor A
[0084] **211** inner bearing surface A
[0085] **212** outer conductor contact spring A
[0086] **219** second connection flange
[0087] **220** dual second outer conductor
[0088] **221** outer bearing surface A
[0089] **222** outer bearing surface B
[0090] **223** outer conductor contact surface A
[0091] **224** outer conductor contact surface B
[0092] **230** first outer conductor B
[0093] **231** inner bearing surface B
[0094] **232** outer conductor contact spring B
[0095] **240** first inner conductor A
[0096] **241** first inner conductor contact
[0097] **242** joint socket A
[0098] **243** center contact surface A
[0099] **244** center contact spring A
[0100] **245** locking means
[0101] **250** dual second inner conductor
[0102] **251** joint ball A
[0103] **252** joint ball B
[0104] **253** joint ball locking means
[0105] **260** first inner conductor B
[0106] **262** joint socket B
[0107] **263** center contact surface B
[0108] **264** center contact spring B
[0109] **271** first spacer
[0110] **272** second spacers
[0111] **273** third spacer
[0112] **300** corner component
[0113] **310** corner outer conductor
[0114] **320** corner inner conductor
[0115] **400** RF connecting system

1. An RF joint for high power coaxial RF lines comprises:
a first outer conductor with a first inner conductor centered therein,
a second outer conductor with a second inner conductor centered therein,
the first inner conductor comprising a joint socket that is mechanically coupled to a joint ball of the second inner conductor and that forms a socket-ball connection, wherein the first inner conductor further comprises a center contact surface that has a shape of a spherical segment and that is
in contact with a center contact spring of the second inner conductor,
wherein:
the first outer conductor has an outer conductor contact spring that is in contact with an outer conductor contact surface at the second outer conductor or
the second outer conductor has an outer conductor contact spring which is in contact with an outer conductor contact surface at the first outer conductor.

2. The RF joint according to claim **1**, wherein the second outer conductor has an outer bearing surface, which is overlapped by an inner bearing surface of the first outer conductor.

3. The RF joint according to claim **1**, comprising a first spacer between the first inner conductor and the first outer conductor.

4. The RF joint according to claim **1**, comprising a second spacer between the second inner conductor and the second outer conductor.

5. The RF joint according to claim **1**, wherein the center contact surface has the shape of the spherical segment with a center point at a center point of the joint ball of the second inner conductor.

6. The RF joint according to claim **1**, wherein the outer conductor contact surface has a hollow spherical shape or a segment thereof, wherein the hollow spherical shape or the segment thereof has a center point at a center point of the joint ball of the second inner conductor.

7. The RF joint according to claim **1**, wherein the RF joint is structured with respect to a center axis of the RF joint that is defined by either a first inner conductor or a second inner conductor and is configured to perform at least one of:
tilting in a plane through the center axis of the RF joint,
tilting orthogonally to the plane through the center axis of the RF joint,
rotating around the center axis of the RF joint,
extending or reducing its length along the center axis of the RF joint.

8. An RF joint component including at least one RF joint according to claim **1**, further comprising at least one of:
an inner conductor contact,
a connection flange, and
an elongated conductor section including corresponding outer and inner conductors.

9. An RF joint component according to claim **8**, wherein the at least one RF joint includes two of said RF joints that are connected in series and mirrored to each other.

10. An RF connecting system including at least two of the RF joint components according to claim **8**, connected together, and further comprising:

at least one straight, curved, or cornered conductor component that includes an outer conductor and an inner conductor centered within the outer conductor.

* * * * *